Uncertainty Quantification Report

Report: UQ-001
Subject: sensor-stream safety margins
Seed: 20260525

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Abstract.

Executive summary: across the featured humanoid sensor stream, live safety margins stayed within the governing gate bounds with the coefficient-of-variation held under its limit; details follow.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords

1 Methodology

Margins are computed against each gate's bound over the sensor stream, with mean and uncertainty bands reported per signal. The method is reproducible from the recorded seed.

2 Margins

Safety margins are summarized below and visualized as bands against the governing bounds.

Figure placeholder (`Images/safety-bands.png`). The float, caption, and `\label` mirror the VVUQ-02 figures; replace the box with `\includegraphics`.

Figure 1: Figure 1. Placeholder safety-band plot across the sensor stream.

3 Margin summary

Signal	Mean margin	Uncertainty band
contact force	within budget	CoV under bound
clearance	above minimum	CoV under bound
dispersion	within bound	band reported

Table 1: Table 1. Per-signal margin summary (modeled on VVUQ-05 Table 1).

4 Sensitivity

The dominant contributor to band width is sensor noise on the clearance channel; tightening it would narrow the bands without changing any gate bound.

Acknowledgments

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Ethical Disclosures

The author declares no competing interests. This is an independent draft and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. All supporting numbers are simulation or illustrative results.

Cite This Report

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Data Availability

Supporting records are openly available in the `kevinkawchak/cancer-automated` and `kevinkawchak/physical-ai-oncology-trials` GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

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